Asset Markets in General Equilibrium

- Next, we model financial assets (rather than state contingent commodities).
- A unit of an asset gives the holder the right to receive some future payment.

- An asset is a title to receive pre-specified amounts of good 1 at date 1.
- An asset is completely characterized by its return vector $\mathbf{r}_k = (r_{k1}, .., r_{kS}) \in \mathbb{R}^S$
  - r_{ks} is the dividend paid to the holder of a unit of r_k if and only if state s occurs.

Definition

The asset return matrix R is an $S \times K$ matrix whose kth column is the return vector of asset k. That is:

$$R = \begin{bmatrix} r_{11} & .. & r_{k1} & .. & r_{K1} \\ .. & .. & .. & .. & .. \\ r_{1s} & .. & r_{ks} & .. & r_{Ks} \\ .. & .. & .. & .. & .. \\ r_{1S} & .. & r_{kS} & .. & r_{KS} \end{bmatrix}$$

A row indicates the returns of all assets in one particular state.

Assets: Examples

Example

An asset that delivers one unit of good 1 in all states:

$$\mathbf{r} = (1, .., 1)$$

If there is only one good, $L = 1$, this the **risk-free** (or safe) asset.

- Why is this not safe with many goods?
  - Because the price of good 1 can change from state to state.
    - An asset matters insofar as it can be transformed into consumption goods; the rate at which one can do this depends on relative prices.

Example

An asset that delivers one unit of good 1 in one state, and zero otherwise:

$$\mathbf{r} = (0, .., 1, .., 0)$$

This is an Arrow security.

Budget Constraints with Asset Markets

Given an asset matrx R, one can define prices and holdings of each asset.

- $\mathbf{q} = (q_1, .., q_K) \in \mathbb{R}^K$ are the asset prices, where q_k is the price of asset k.
- $\mathbf{z}_i = (z_{1i}, .., z_{Ki}) \in \mathbb{R}^K$ are consumer i's holdings of each asset.
  - This is called a portfolio: it shows how many units of each asset i owns.

Assets are traded at time 0, while returns are realized at time 1. At that time, agents decide how much to consume and they trade in spot markets.

- As in Radner, i's budget constraints are

$$\underbrace{\sum_{k=1}^{K} q_k z_{ki} \leq 0}_{\text{time 0}} \quad \text{and} \quad \underbrace{\mathbf{p}_s \cdot \mathbf{x}_{si} \leq \mathbf{p}_s \cdot \boldsymbol{\omega}_{si} + \sum_{k=1}^{K} p_{1s} z_{ki} r_{sk}}_{\text{time 1}} \quad \text{for each } s$$

- Income at time 1 is given by the value of endowment plus the income one obtains by selling the returns of the assets one owns.
  - As usual, one can normalize the spot price of good one to be 1.

Assets vs State-contingent Commodity in Radner

Question

What is the difference between these budget constraints

$$\sum_{k=1}^{K} q_k z_{ki} \leq 0 \qquad \text{and} \qquad \mathbf{p}_s \cdot \mathbf{x}_{si} \leq \mathbf{p}_s \cdot \boldsymbol{\omega}_{si} + \sum_{k=1}^{K} p_{1s} z_{ki} r_{sk} \qquad \text{for each } s$$

and the ones in a Radner equilibrium?

- Here, assets' dividends are given; one focuses only on the portfolio choice $\mathbf{z}_i$.
- In Radner, the dividends are 'constructed' by the consumers' choice of trades in the state-contingent commodity.
- Formally, in Radner one implicitly assumes S different assets, each with returns $r_s = 1$ in state s and zero otherwise.
- If $S = K$, we can write Radner using the z and r above:

$$z_s^{Radner} = \sum_{s=1}^{S} z_s r_s.$$

Budget Set

What do budget sets look like?

- Normalizing $p_{1s} = 1$, one can write i's budget set as

$$B_i(\mathbf{p}, \mathbf{q}, R) = \left\{ \mathbf{x}_i \in \mathbb{R}^{LS}_+ : \begin{array}{c} \text{there is } \mathbf{z}_i \in \mathbb{R}^K \\ \text{such that} \end{array} \begin{array}{c} \mathbf{q} \cdot \mathbf{z}_i \leq 0 \\ \text{and} \\ \mathbf{p} \cdot (\mathbf{x}_i - \boldsymbol{\omega}_i) \leq R\mathbf{z}_i \end{array} \right\}$$

- The second part of the budget set defines S inequalities

$$\begin{array}{rcl} \mathbf{p}_1 \cdot (\mathbf{x}_{1i} - \boldsymbol{\omega}_{1i}) & \leq & z_{1i} r_{11} + ... + z_{ki} r_{k1} + ... + z_{Ki} r_{K1} \\ ... & & ... \\ \mathbf{p}_s \cdot (\mathbf{x}_{si} - \boldsymbol{\omega}_{si}) & \leq & z_{1i} r_{1s} + ... + z_{ki} r_{ks} + ... + z_{Ki} r_{Ks} \\ ... & & ... \\ \mathbf{p}_S \cdot (\mathbf{x}_{Si} - \boldsymbol{\omega}_{Si}) & \leq & z_{1i} r_{1S} + ... + z_{ki} r_{kS} + ... + z_{Ki} r_{KS} \end{array}$$

  where each $\mathbf{p}_s \in \mathbb{R}^L$

- $R\mathbf{z}_i$ is the consumer's 'financial income' from choosing the portfolio $\mathbf{z}_i$.
- Given R, each value of $R\mathbf{z}_i$ determines a possible income available to the consumer.
  - Different $\mathbf{z}_i$s change the consumer's future income across states.

Radner Equilibrium with Asset Markets

Definition

A Radner equilibrium with assets is given by asset prices $\mathbf{q}^* \in \mathbb{R}^K$, prices $\mathbf{p}_s^* \in \mathbb{R}^L$ in each state s, portfolios $\mathbf{z}_i^* \in \mathbb{R}^K$, and consumption $\mathbf{x}_{si}^* \in \mathbb{R}^L$ for each s such that:

1. for each i, $\mathbf{z}_i^*$ and $\mathbf{x}_i^*$ solve
$$\max_{\mathbf{z}_i \in \mathbb{R}^K, \mathbf{x}_i \in \mathbb{R}_+^{S \times L}} U_i(\mathbf{x}_i)$$
subject to
$$\sum_{k=1}^{K} q_k^* z_{ki} \leq 0 \quad \text{and} \quad \mathbf{p}_s^* \cdot \mathbf{x}_{si} \leq \mathbf{p}_s^* \cdot \boldsymbol{\omega}_{si} + \sum_{k=1}^{K} p_{1s}^* z_{ki} r_{sk}$$
2. all markets clear; that is:
$$\sum_{i=1}^{I} z_{ki}^* \leq 0 \quad \text{and} \quad \sum_{i=1}^{I} \mathbf{x}_{si}^* \leq \sum_{i=1}^{I} \boldsymbol{\omega}_{si} \quad \text{for all } s \text{ and } k$$

- The definition is familiar, but there are two new objects: the optimal portfolio and the equilibrium assets' prices.
- What can one say about them? What do we know about $\mathbf{q}^*$?

Complete Markets

Definition

An asset structure R is complete if *rank* $R = S$.

Example

There are S different Arrow securities; then

$$R = \begin{bmatrix} 1 & 0 & .. & .. & 0 \\ 0 & 1 & 0 & .. & 0 \\ 0 & 0 & 1 & .. & 0 \\ .. & .. & .. & .. & .. \\ 0 & .. & .. & 0 & 1 \end{bmatrix}$$

which is the identity matrix and therefore has rank S.

Example

There are three states, so $S = 3$, and $R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ which has full rank.

Arbitrage

Arbitrage

An abitrage opportunity is the possibility to make strictly positive profits at no risk.

- Under what conditions such opportunities do not exist?

Definition

Asset prices $\hat{\mathbf{q}} \in \mathbb{R}^K$ satisfy absence of arbitrage if there is no $\mathbf{z} \in \mathbb{R}^K$ such that

$$\hat{\mathbf{q}} \cdot \mathbf{z} \leq 0, \qquad R\mathbf{z} \geq 0 \qquad \text{and} \qquad R\mathbf{z} \neq 0.$$

- $\hat{\mathbf{q}}$ are sometimes called no-arbitrage prices.

- This can be interpreted as

$$\underbrace{\hat{\mathbf{q}} \cdot \mathbf{z} \leq 0}_{\text{affordable portfolio}} \quad , \quad \underbrace{R\mathbf{z} \geq 0 \qquad \text{and} \qquad R\mathbf{z} \neq 0}_{\text{strictly positive profits for sure}}$$

- No arbitrage means there does not exists an affordable portfolio that yields non-negative returns in all states and strictly positive returns in some state.
- Preferences are nowhere to be seen in this definition.

Arbitrage and Asset Prices

- No arbitrage imposes restrictions on prices.

No arbitrage and prices

Let

$$R = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \qquad \text{so that} \qquad Rz = \begin{bmatrix} z_1 + z_2 + z_3 \\ z_1 + z_2 \\ z_1 \end{bmatrix}$$

- Here, no arbitrage implies that if $z_1 > 0$ then $q_1 > 0$. Why?
  - Can always pick z_2 and z_3 equal to zero.

- This result generalizes beyond the example.

No Arbitrage and Equilibrium Asset Prices

Theorem

Assume $r_k \geq 0$ and $r_k \neq 0$ for all k (return vectors are nonnegative and non zero), and suppose preferences are strictly monotone.
Then, any $\mathbf{q}^* \in \mathbb{R}^K$ that is part of a Radner equilibrium satisfies no-arbitrage.

- Under simple restrictions on assets and preferences, equilibrium implies absence of arbitrage.
- Intuitively, arbitrage implies some money is left on the table.
- Given the efficiency properties of equilibirum, this should not happen.
- This is similar in spirit to the First Welfare Theorem.
- Prove this as an exercise.

Equilibrium and Linear Asset Prices

Theorem (Martingale Pricing)

Assume $r_k \geq 0$ and $r_k \neq 0$ for all k (return vectors are nonnegative and non zero), and suppose preferences are strictly monotone. Then, for any $\mathbf{q}^* \in \mathbb{R}^K$ that is part of a Radner equilibrium, we can find non-negative numbers $\mu_s \geq 0$ which satisfy

$$q_k^* = \sum_{s=1}^{S} \mu_s r_{sk} \text{ for all } k$$

What does this equality imply?

- Prices are a linear combination of dividends: $[\mathbf{q}^*]^T = \boldsymbol{\mu} \cdot R$
- Suppose asset k pays a constant amount in all states: $r_{sk} = c$ for all s.
- Normalize the price of this asset to be c, so that $c = \sum_{s=1}^{S} \mu_s c$;then $1 = \sum_{s=1}^{S} \mu_s$
- Thus, equilibrium asset prices must equal an expected value of their returns.
- The 'probability' used to compute the expected value is the same for all assets.

No-Arbitrage Implies Linear Asset Prices

- Since we have already shown that equilibrium implies no-arbitrage, to prove the Martingale Pricing Theorem we only need to prove the following lemma.

Lemma

Suppose $r_k \geq 0$ and $r_k \neq 0$ for all k (return vectors are nonnegative and non zero). If asset prices $\mathbf{q} \in \mathbb{R}^K$ satisfy the no-arbitrage condition, then there exists a vector $\boldsymbol{\mu} \in \mathbb{R}^S_+$ such that

$$\mathbf{q}^T = \boldsymbol{\mu} \cdot R$$

- The proof follows from convexity of the set of no arbitrage portfolios.
  - The main step uses the separating hyperplane theorem, and should remind you of earlier proofs:
    - Separate the positive orthant from the set of incomes that can be obtained given a no-arbitrage (equilibirum) price vector.
- This is also left as an exercise.
- Notice that all we need for linear pricing is no-arbitrage.
- There are no restrictions on preferences for the Lemma!
- The connection with equilibrium is made by adding strictly monotone preferences.

Implications of Linear Pricing

Suppose there is an asset that delivers one unit of good 1 in all states:

$$\mathbf{r}_1 = (1, .., 1)$$

- Let $\mathbf{q}$ be a no-arbitrage price vector and normalize $q_1 = 1$.
- If $\boldsymbol{\mu}$ satisfies $\mathbf{q}^T = \boldsymbol{\mu} \cdot R$, we have

$$\boldsymbol{\mu} = (\mu_1, .., \mu_S) \geq 0 \qquad \text{and} \qquad \sum_{s=1}^{S} \mu_s = \boldsymbol{\mu} \cdot r_1 = q_1 = 1$$

- Hence, for all assets different from asset 1,

$$q_k = \sum_{s=1}^{S} \mu_s r_{sk} \geq \min_s r_{sk} \qquad \text{and} \qquad q_k = \sum_{s=1}^{S} \mu_s r_{sk} \leq \max_s r_{sk}$$

- Intuitively, prices must be between the lowest and highest dividend.

Next Class

- Talk about the exam: it will be next Monday at 9:30 in this room and last 3 hours.
- Have some fun with Luca's research (this material will not be on the final).
- Material from today will also not be on the final.